Chidi Nna

Data Engineer/Scientist/Analyst

WORK EXPERIENCE

BenefitScape, Natick MA 06/2025– Present

ACA Data Analyst/Manager (Full Time)

- Designed and optimized FME data workflows to process healthcare datasets for over **600** clients, ensuring accuracy, efficiency, and compliance with Affordable Care Act (ACA) reporting requirements.
- Developed and refined custom FME transformers to streamline repetitive tasks, accelerate processing times, and enhance workflow clarity for other team members.
- Improved data management processes by revising and standardizing FME scripts, reducing reporting cycle times by **25%** and enabling the processing of healthcare datasets for clients in compliance with ACA requirements.
- Built ETL pipelines in FME to load cleaned and validated data for use in graphs, dashboards, and statistical analysis, enabling year-over-year comparisons and insights for clients by processing **20,000**+ records across multiple Excel spreadsheets.
- Enabled Python startup and shutdown scripts within FME workspaces to automate file movement, cleanup operations, and other preprocessing/postprocessing tasks, improving automation and efficiency for specific client workflows

Bureau Of Land Management, Remote 05/2024 – 10/2024

Data Scientist/Analyst (Full Time- Contractor)

- Designed and implemented an ontology database system using TopBraid EDG, integrating GraphQL to efficiently query and manage geospatial data from the Bureau of Land Management, ranging from **2MBs** to **3GBs** in size. This improved data organization and visualization for complex datasets
- Automated data mapping processes in TopBraid EDG using JavaScript and Swagger UI, enabling seamless integration of Excel data into structured database
- Collaborated with different government teams to build comprehensive reports and dashboards using visualization tools for informed decision-making.
- Reviewed and analyzed data standards reports to recommend the best practices for data integration and ensure compliance with Bureau of Land Management guidelines.

Bryant University, Smithfield, RI

Undergraduate Researcher (Part Time) 06/2022 – 09/2022

- Scraped and standardized citation data from Publish or Perish, consolidating **2,000+** publication records across multiple Bryant professors into a relational database for institutional use."
- Extracted, transformed, and loaded (ETL) publication data from Hazard Publish and Perish using SQL and Python, building a structured database to facilitate university research.

SKILLS

Programming Languages:

Expert: **Scala, Python, SQL, Java**

Proficient: **C, C++, R**

Hard Skills:

- Data Analysis
- Data Visualization
- Data Wrangling
- Database Skills

Tools:

- Tableau, Power BI, Microsoft Excel
- Xcode, Firebase
- MySQL, JupyerNotebook, Google Colab
- AWS S3, Microsoft Azure
- Pyspark
- Xcode, Firebase
- Oracle Database, Databricks
- MongoDB
- PyTorch, TensorFlow, Docker, Kubernetes
- Apache Airflow, Git

***Techniques*:**

- Classification
- Regression
- Clustering
- Pandas, NumPy, Scikit-Learn, Matplotlib
- Text Analytics
- Univariate/Bivariate Analysis
- Text & Descriptive Analytics

EDUCATION

University of New Haven,

Master Of Science,

Computer Science

West Haven, CT - 2025

Bryant University,

Bachelor of Science,

Data Science

Minor in Business Administration

Smithfield, RI -2023

Awards: Dean List

COURSE WORK

Data Visualization, Database Management, Information Technology and Analysis, Statistics II, Data Warehouse II, Big Data Analytics, Introduction to Machine Learning, Cloud Computing, Object-Oriented Programing, Data Structures. Distributed Database Systems, Advanced Database Systems. Data Mining

OTHER

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Phone #: 857-352-8425
GitHub: https://github.com/cnna4
Credentials: PIV Card
Hiring Authorities: Direct Hire Authority

- Partnered with faculty to validate data accuracy, integrity ensuring the reliability of research outputs and alignment with project goals. Created interactive Tableau dashboards and detailed reports to analyze and present publication data,

Google Code U, Remote

Software Engineering (Intern) 02/2021 – 05/2021

- Collaborated with a team of four student engineers in a remote internship to design and implement backend systems for an internal application aimed at optimizing task management and workflow efficiency.
- Developed and maintained scalable Java-based backend systems, focusing on efficient data processing, storage, and retrieval to support application performance and reliability.
- Built and maintained data pipelines for extracting, transforming, and loading (ETL) data, leveraging Java to process large datasets and improve data accessibility for downstream applications.
- Conducted thorough testing, debugging, and performance tuning of data pipelines and backend systems to ensure high availability, accuracy, and system reliability.
- Utilized Git for version control and collaborated with team members through pull requests and code reviews, to be complying with coding standards and best practices.

University of New Haven, West Haven 06 /2024 – 06/2025

Graduate Assistant (Part Time)

- Demonstrated leadership as a Graduate Assistant by selling over **8,000** tickets to campus events for graduate students, co-organizing numerous activities, actively engaging in initiatives to enhance student life, and serving on the Board of Governors to provide student perspectives on academic policies and curriculum development
- Developed and distributed Qualtrics surveys to collect student feedback, using the platform's built in graphing tools to generate visual reports and identify key trends in student satisfaction and preferences.
- Analyzed survey data from Qualtrics to evaluate the success of campus events, using statistical insights and visualizations to recommend improvements for future planning.

PROJECTS

Product Popularity Analysis on Amazon (Python)

- Conducted in-depth analysis of Amazon product data, using univariate and bivariate analysis for exploration, F1 score, MSE to measure accuracy, and clustering to identify patterns driving product popularity. Utilized clustering techniques such as K-Means to segment products into meaningful groups based on features such as ratings, and pricing, uncovering unseen patterns in customers preferences

Solar Panel Dust Detection using Deep Learning Models for Prototype Vacuum Dust Detection

- Conducted image classification research using a Kaggle dataset of ~**2,500** solar panel images (clean vs. dusty) to explore the impact of dust on solar efficiency, using scaling, resizing, and augmentation techniques to improve model robustness. Find full project Here
- Implemented and tested three deep learning architectures — Custom CNN, MobileNetV2, and ResNet50 — comparing their performance in accuracy, recall, and F1-score.
- Identified ResNet50 as the strongest model for capturing true positives on dusty panels, while also highlighting the pros and cons of MobileNetV2 and Custom CNN.
- Evaluated overfitting/underfitting trends across models and recommended improvements for dataset balance and real-world deployment.